

The Cooper Union Department of Electrical Engineering

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ECE478 Financial Signal Processing

Problem Set I

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1. **Be a Financial Engineer!**

Here we will study mixes of various derivatives based on an underlying security S_T , with the property that the payout at maturity date T depends only on the final value of the security, i.e., the payout has the form $V(S_T)$.

Holding a long position means the value in the portfolio at T is $+V(S_T)$, and a short position means the value is $-V(S_T)$.

Given a strike price K , define European call and put options, respectively, as:

$$\begin{aligned}c(T, T) &= (S_T - K)^+ \\p(T, T) &= (K - S_T)^+\end{aligned}$$

Define a *digital* call or put option to have respective payout as follows:

$$\begin{aligned}d_C(T, T) &= 1 \text{ if } S_T > K, 0 \text{ otherwise} \\d_p(T, T) &= 1 \text{ if } S_T < K, 0 \text{ otherwise}\end{aligned}$$

Combining various amounts of long and short positions in call and put options can yield a general piecewise linear but overall continuous value $V(S_T)$ at the time of maturity. Including digital options allows discontinuities. Here we will see several important combinations.

Use a computer to generate plots by assigning reasonable values to the parameters K , etc. You just need to produce one example of each. In all cases, horizontal axis is $S_T > 0$ and the vertical axis is $V(S_T)$.

First two comments:

- As you graph these, keep in mind you are looking at the “payout” $V(S_T)$. If you purchase one of these derivatives at a price say V_t at t , then your net “profit” or “loss” at time T would be $V(S_T) - V_t/Z(t, T)$ (i.e., subtract off the money you would have had in your pocket if you never purchased the derivative). This would be a more useful graph, perhaps, e.g., to determine the range of S_T for which you end up with a net positive, but that would require determining the price at t , V_t , which we are not dealing with here. So, we will create these graphs ignoring the purchase price of the portfolio—we are just looking at final payout, not net profit/loss.
- One thing to look for in any of these instruments: is the potential loss unbounded? In other words, is $\min V(S_T)$ finite?

Also, in what follows below, except for one case I'm not asking you to derive an analytic formula for $V(S_T)$. If you write core code to compute the payout for the call, put, digital call, digital put, then just write code to combine these as suggested to obtain the graphs.

- (a) Graph $V(S_T)$ for each of the following: long call, long put, short call, short put.
- (b) A *straddle* is a combination of one unit of a call and one unit of a put for the same security at the same strike price K . Verify $V(S_T) = |S_T - K|$, and graph this for one choice of K .
- (c) A *call-put spread* is to long a call at K_1 and short a call at K_2 with $K_1 < K_2$. Graph $V(S_T)$ for one choice of the pair K_1, K_2 .
- (d) A *butterfly* is a combination of the following. Let $K_1 < K_2$, and $0 < \lambda < 1$. Let $K^* = \lambda K_1 + (1 - \lambda) K_2$. Then:
 - long λ calls at strike price K_1
 - long $(1 - \lambda)$ calls at strike price K_2 .
 - short 1 call at strike price K^* .

Fix one choice for $K_1 < K_2$. Graph a butterfly $V(S_T)$ for each of the following cases: $\lambda = 1/3, 1/2, 2/3$.

- (e) *Call ladder*: with $K_1 < K_2 < K_3$, long one K_1 call, short one K_2 call, short one K_3 call.
- (f) A *digital call spread*: with $K_1 < K_2$, a long a digital call with strike K_1 and short a digital call with strike K_2 .

2. Radon-Nikodym Derivative for Gaussian

We want to examine the Radon-Nikodym derivative that shifts the mean for a general Gaussian to 0. Specifically, it should map P , under which $X \underset{P}{\sim} N(\mu, \sigma^2)$, to $\tilde{P}$, under which $X \underset{\tilde{P}}{\sim} N(0, \sigma^2)$. The proposed Radon-Nikodym derivative has the form:

$$Z = e^{cX+d}$$

for constants c, d to be determined. The method in the lecture notes used the mgf. Here, we will take a different approach. We are going to match two formulas for:

$$\tilde{P}(a < X < b)$$

- (a) First, express this in the form of a $\tilde{P}$ -expectation of a random variable Y , i.e., specify Y such that:

$$\tilde{P}(a < X < b) = E_{\tilde{P}}(Y)$$

- (b) Starting with $X \underset{P}{\sim} N(\mu, \sigma^2)$, use the Radon-Nikodym Theorem to find an integral formula for $E_{\tilde{P}}(Y)$, using Z as above
- (c) Find c and d so that the integral you get matches what you would get for $\tilde{P}(a < X < b)$ if $X \underset{\tilde{P}}{\sim} N(0, \sigma^2)$.
- (d) Write your final formula for Z .
- (e) We clearly have $Z > 0$ a.s. But to verify Z is a valid Radon-Nikodym derivative, we also have to check that $E_P(Z) = 1$. Use the mgf of Gaussian (formula given in the notes) to verify this!